
Two-Pass Reconstruction-Error-Guided Attention for Adversarial Autoencoder-Based Network Anomaly Detection

Research Proposal

Deep Learning Project — Network Anomaly Detection

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Abstract

Network intrusion detection requires identifying rare, unseen attack patterns in high-dimensional traffic data with minimal or no labels. Established unsupervised approaches on the KDD99 benchmark—deep autoencoders [Zavrak & Iskefiyeli, 2021] and GAN-based discriminator scoring [Zenati et al., 2018]—share two structural limitations: all input features are treated uniformly regardless of their discriminative relevance, and the latent representation is unconstrained, making anomaly reasoning in latent space ill-defined.

This proposal describes an architectural experiment that addresses both limitations through a single novel mechanism: **Two-Pass Reconstruction-Error-Guided Attention (RE-Attention)**. In the first pass, a standard encoder-decoder reconstructs the input; the per-feature reconstruction error vector is then used as the attention signal to re-weight the input for a second encoding pass whose latent code is regularised by an adversarial Gaussian prior. This creates a self-corrective feedback loop in which the model identifies its own reconstruction failures and attends to them, enabling a richer and more targeted anomaly representation than input-driven or bottleneck-driven attention reported in prior work [Al-Qatf et al., 2021, Abdulganiyu et al., 2025]. A controlled five-condition ablation study

isolates the contribution of each component, and the attention weights are visualised per attack family to provide interpretable insights into which network traffic features most strongly signal each class of intrusion.

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1. Introduction

1.1. Background and Motivation

Network intrusion detection is one of the most studied applications of machine learning in cybersecurity. The core challenge is that in real-world networks, attack traffic represents a tiny fraction of total traffic—often less than 1%—and many attacks, particularly zero-day exploits, have never been seen before. This makes supervised classification insufficient as a sole defence: a model that has only seen known attacks will fail to detect novel ones.

Unsupervised anomaly detection addresses this by learning only what *normal* traffic looks like, then flagging anything that deviates as potentially malicious. The fundamental insight is that a model trained exclusively on normal data will fail to reconstruct or generate anomalous samples well, and this failure itself becomes the detection signal.

1.2. Baseline Approaches and Their Limitations

The two dominant unsupervised paradigms for network anomaly detection on KDD99 are reconstruction-based autoencoders [Zavtrak & Iskefiyeli, 2021] and generative adversarial networks [Zenati et al., 2018]. Both exhibit structural weaknesses that motivate this proposal:

1. **Unstructured latent space (AE):** Standard autoencoders impose no constraint on the bottleneck representation [Zavtrak & Iskefiyeli, 2021]. A 4-dimensional latent code can take any shape; there is no principle by which a latent code can be declared “normal-looking” without re-decoding through the full decoder.
2. **Training instability (GAN):** GANs operating on the full 114-dimensional input space are prone to convergence failure [Zenati et al., 2018]. The discriminator is trained to separate *generated* from *real* samples—a proxy for normality, not a direct measure of it.
3. **Uniform feature treatment (both):** All 114 features receive equal weight during encoding. In KDD99, many one-hot encoded service and flag features are near-zero for most samples, while features such as `error_rate` and `dst_host_count` are highly discriminative for specific attack families. No existing approach on this dataset learns per-feature importance adaptively from the model’s own reconstruction behaviour.

1.3. Proposed Contribution

We propose a new architectural mechanism—**Two-Pass Reconstruction-Error-Guided Attention (RE-Attention)**—embedded in an Adversarial Autoencoder framework. A literature search (Section 3) confirms that while adversarial AEs, attention AEs, and dual scoring have all been explored separately and in partial combinations, the specific mechanism of using the per-feature reconstruction error vector as the attention signal for a second encoding pass has not been reported in any prior work on network anomaly detection.

2. Dataset Description

2.1. KDD Cup 1999 (KDD99)

The KDD Cup 1999 dataset [Tavallaee et al., 2009] was created from the DARPA 1998 network intrusion dataset and contains approximately 5 million network connection records. Each record describes a single TCP/IP connection with 41 features: 34 continuous (traffic statistics such as byte counts, error rates, and connection durations) and 7 categorical (protocol type, service, connection flag, and binary indicators).

After one-hot encoding the categorical variables, the feature dimension expands to **114 features**. The dataset contains 23 distinct attack types grouped into four families:

Table 1: KDD99 Attack Families and Representative Types

Family	Description	Examples
DoS	Denial-of-service; overwhelm the target server	neptune, smurf, back, teardrop
Probe	Silent network scanning; reconnaissance before attack	ipsweep, nmap, satan, portsweep
R2L	Remote-to-local; attempt unauthorised access from network	warezclient, guess_passwd, ftp_write
U2R	User-to-root; attempt to escalate privileges to root	rootkit, buffer_overflow, loadmodule

Following standard experimental practice in this domain [Zavtrak & Iskefiyeli, 2021, Zenati et al., 2018], the dataset is subsampled to contain only 1% anomalies relative to normal traffic, simulating the realistic imbalance encountered in operational networks. The pre-processing pipeline one-hot encodes categorical variables, applies **MinMaxScaler** normalisation, and splits the data 75/25 into training and test sets.

2.2. NSL-KDD

The NSL-KDD dataset [Tavallaee et al., 2009] addresses known deficiencies in KDD99: duplicate records that cause classifiers to over-fit to common patterns are removed, and the selection of records from each difficulty level is inversely proportional to their prevalence in the original dataset, making rare attack families harder to detect. It serves as the secondary evaluation benchmark in this study, consistent with its use in prior work [Abdulganiyu et al., 2025, Zavtrak & Iskefiyeli, 2021].

3. Related Work

The literature on deep learning for network anomaly detection is extensive. We focus on the work most relevant to the proposed architecture, organised by the primary technique used.

3.1. Autoencoders for Network Intrusion Detection

Autoencoders have been the dominant deep learning approach for unsupervised network anomaly detection. Zavtrak & Iskefiyeli [2021] provide a systematic comparison of vanilla AE, denoising AE, sparse AE, and variational AE on the KDD99 and NSL-KDD datasets,

establishing that reconstruction error is a reliable anomaly signal when training is restricted to normal traffic. ? specifically improve standard AE performance on NSL-KDD through architectural choices, providing an upper bound for vanilla AE baselines.

Relevance to this work: These papers establish the AE baseline (our Condition 1) and confirm the reconstruction-error paradigm on the exact datasets used in this study.

3.2. Adversarial Regularisation of Autoencoders

A key limitation of vanilla AEs is the absence of any constraint on the latent space. Two papers address this for general anomaly detection:

Zenati et al. [2018] propose **Adversarially Learned Anomaly Detection (ALAD)**, a bi-directional GAN framework where an encoder maps data to latent codes simultaneously with the generator mapping latent codes to data. Two discriminators enforce cycle-consistency in both the data space and the latent space. ALAD uses dual anomaly scores based on reconstruction error and discriminator outputs, achieving state-of-the-art results on KDD99 and several image datasets with 454 citations as of 2026. ALAD is the closest prior work to the adversarial regularisation component we propose.

Lunardi et al. [2022] present **ARCADE**, an adversarially regularised convolutional AE specifically for network anomaly detection. ARCADE is trained exclusively on normal traffic and uses an adversarial training strategy to reduce the AE’s ability to reconstruct out-of-distribution flows, evaluated on the CICIDS2017 dataset. Crucially, ARCADE uses no attention mechanism and no dual anomaly scoring.

Puuska et al. [2018] combine Haar wavelet transforms with an adversarial autoencoder for detecting malicious TLS-encrypted network traffic, demonstrating that adversarial AE consistently outperforms vanilla AE in the network security domain.

Gap: ALAD, ARCADE, and the wavelet-AAE work all use input-driven or data-space adversarial training. None employ attention guided by reconstruction error.

3.3. Attention Mechanisms for Intrusion Detection

Attention mechanisms have been integrated with autoencoders and sequence models for network intrusion detection, but almost exclusively as temporal attention over LSTM hidden states.

Abdulganiyu et al. [2025] propose a multi-model architecture combining a class-wise focal-loss variational AE (**CWFL-VAE**) with an attention mechanism-based LSTM (**AM-LSTM**) for imbalanced network traffic on NSL-KDD and CSE-CIC-IDS2018. The attention is temporal (over LSTM time steps), not feature-level, and the VAE is used for data balancing rather than direct anomaly scoring.

Al-Qatf et al. [2021] embed an attention module at the bottleneck of an AE to selectively enhance important dimensions of the latent vector before decoding. The attention is applied to the *latent code*, not the input features, and is not conditioned on reconstruction error.

Gap: Existing attention approaches for network intrusion are either (a) temporal attention over sequence models, or (b) bottleneck attention on the latent code. **No prior work uses the per-feature reconstruction error vector as the attention signal for a second encoding pass.**

3.4. Combined and Hybrid Approaches

Benabderrahmane et al. [2025] propose the most comprehensive hybrid to date: **AL-ADAEN**, an Active Learning-assisted Attention Adversarial Dual AutoEncoder. The framework combines active learning (oracle querying for uncertain samples), an adversarial dual autoencoder, and attention mechanisms for Advanced Persistent Threat (APT) detection on DARPA Transparent Computing provenance traces (not KDD99). While structurally ambitious, ALADAEN’s active learning loop requires an oracle and is evaluated on a different domain. The attention mechanism is not specified as error-driven.

Xu et al. [2024] combine a VAE and Wasserstein GAN into a **VAE-WGAN** hybrid for network intrusion detection on KDD99. The generative component is used primarily for data augmentation of minority attack classes, and the anomaly detection uses a downstream LSTM-MSCNN classifier rather than the latent representation directly.

3.5. Summary of Gaps

Table 2 summarises the key prior works and identifies the architectural features missing from each that motivate this proposal.

Table 2: Comparison of Related Works and the Proposed Method

Method	Year	Adv. Latent	Attention	Dual Score	KDD99/NSL
Vanilla AE [Zavrak & Iskefiyeli, 2021]	2021	×	×	×	✓
ALAD [Zenati et al., 2018]	2018	✓	×	✓	✓
ARCADE [Lunardi et al., 2022]	2022	✓	×	×	×
Wavelet+AAE [Puuska et al., 2018]	2018	✓	×	×	×
VAE+Attn-LSTM [Abdulganiyu et al., 2025]	2025	×	Temporal	×	✓
Attn-AE (bottleneck) [Al-Qatf et al., 2021]	2021	×	Latent	×	×
ALADAEN [Benabderrahmane et al., 2025]	2025	✓	Unspecified	✓	×
VAE-WGAN [Xu et al., 2024]	2024	✓	×	×	✓
Proposed (RE-Attn AAE)	2026	✓	Error-guided	✓	✓

4. Problem Formulation

Let $\mathbf{x} \in \mathbb{R}^d$ denote a network connection feature vector ($d = 114$ after one-hot encoding). The training set $\mathcal{D}_{\text{train}} = \{\mathbf{x}_i\}_{i=1}^N$ contains predominantly normal traffic (99% normal, 1% anomalous). The test set $\mathcal{D}_{\text{test}}$ contains an unknown mixture of normal and anomalous connections with ground-truth binary labels $y_i \in \{0 \text{ (normal)}, 1 \text{ (anomaly)}\}$.

The objective is to learn a function $f : \mathbb{R}^d \rightarrow \mathbb{R}_{\geq 0}$ that assigns an anomaly score $s_i = f(\mathbf{x}_i)$ such that a threshold τ maximises the area under the ROC curve (AUC-ROC) and the F1-score on $\mathcal{D}_{\text{test}}$.

Key Challenges:

1. *Class imbalance*: anomalies comprise $\leq 1\%$ of traffic.
2. *Feature heterogeneity*: 114 features vary greatly in discriminative power; uniform weighting dilutes the signal from informative features.
3. *Latent space structure*: without a prior on the latent distribution, anomaly scoring in latent space is ill-defined.

4. *Multi-family attacks*: four attack families (DoS, Probe, R2L, U2R) have distinct signatures; a single threshold may fail on the rarer families (U2R, R2L).

5. Proposed Methodology

5.1. Overview

We propose the **RE-Attention Adversarial Autoencoder (RE-Attn-AAE)**, an architecture built on three components that address the four challenges stated above:

1. **Two-Pass Reconstruction-Error-Guided Attention (RE-Attention)**: addresses feature heterogeneity by computing attention weights from where the model is currently failing, not from the raw input.
2. **Adversarial Gaussian Latent Regularisation**: addresses latent space structure by enforcing $\mathbf{z} \approx \mathcal{N}(0, I)$ through a small latent discriminator.
3. **Dual Anomaly Scoring with learnable blend**: addresses class imbalance and multi-family attacks by combining two complementary signals: reconstruction error and latent normality.

5.2. Architecture

5.2.1 Pass 1 — Standard Reconstruction

The first pass is a standard deep autoencoder following the architecture established for KDD99 in Zavrak & Iskefiyeli [2021]:

$$\mathbf{z}^{(1)} = \text{Enc}^{(1)}(\mathbf{x}) \in \mathbb{R}^4 \quad (1)$$

$$\hat{\mathbf{x}} = \text{Dec}(\mathbf{z}^{(1)}) \in \mathbb{R}^{114} \quad (2)$$

$$\mathbf{e} = (\mathbf{x} - \hat{\mathbf{x}})^2 \in \mathbb{R}^{114} \quad (3)$$

where $\text{Enc}^{(1)}$ has layers $114 \rightarrow 96 \rightarrow 64 \rightarrow 48 \rightarrow 16 \rightarrow 4$ (all tanh, with 10% dropout), and Dec is its mirror image $4 \rightarrow 16 \rightarrow 48 \rightarrow 64 \rightarrow 96 \rightarrow 114$.

5.2.2 RE-Attention Layer

The per-feature squared reconstruction error $\mathbf{e}$ is passed through a small two-layer network to produce a soft attention mask:

$$\mathbf{a} = \sigma(W_2 \cdot \text{ReLU}(W_1 \mathbf{e} + \mathbf{b}_1) + \mathbf{b}_2) \in [0, 1]^{114} \quad (4)$$

where $W_1 \in \mathbb{R}^{64 \times 114}$, $W_2 \in \mathbb{R}^{114 \times 64}$, and σ is the sigmoid function. The attended input is:

$$\tilde{\mathbf{x}} = \mathbf{x} \odot \mathbf{a} \quad (5)$$

Interpretation: A high reconstruction error on feature j in Pass 1 produces a large e_j , which pushes a_j toward 1, preserving that feature in $\tilde{\mathbf{x}}$ for the second pass. Features that are already well-reconstructed (low e_j) are attenuated. The model learns *where it is failing* and directs encoding capacity there.

5.2.3 Pass 2 — Adversarially Regularised Re-Encoding

The attended input $\tilde{\mathbf{x}}$ is encoded by a second encoder of the same architecture to produce the refined latent code:

$$\mathbf{z}^{(2)} = \text{Enc}^{(2)}(\tilde{\mathbf{x}}) \in \mathbb{R}^4 \quad (6)$$

A small latent discriminator D_{lat} enforces a Gaussian prior on $\mathbf{z}^{(2)}$:

$$D_{\text{lat}}(\mathbf{z}) = \sigma(W_3 \cdot \text{ReLU}(W_2' \cdot \text{ReLU}(W_1' \mathbf{z}))) : \mathbb{R}^4 \rightarrow [0, 1] \quad (7)$$

where the layers are $4 \rightarrow 32 \rightarrow 16 \rightarrow 1$ (a small and stable network given the 4-dimensional input).

5.3. Training Procedure

Training alternates between three phases per mini-batch:

Algorithm 1: RE-Attn-AAE Training (one mini-batch)

Input: Batch $\{\mathbf{x}_i\}_{i=1}^B$ from $\mathcal{D}_{\text{train}}$

Phase 1 — Reconstruction

1. $\hat{\mathbf{x}} \leftarrow \text{Dec}(\text{Enc}^{(1)}(\mathbf{x}))$
2. $\mathbf{e} \leftarrow (\mathbf{x} - \hat{\mathbf{x}})^2$ *// per-feature squared error*
3. $\mathbf{a} \leftarrow \text{RE-Attn}(\mathbf{e}); \tilde{\mathbf{x}} \leftarrow \mathbf{x} \odot \mathbf{a}$
4. $\hat{\mathbf{x}}^{(2)} \leftarrow \text{Dec}(\text{Enc}^{(2)}(\tilde{\mathbf{x}}))$
5. $\mathcal{L}_{\text{recon}} \leftarrow \|\mathbf{x} - \hat{\mathbf{x}}\|^2 + \|\mathbf{x} - \hat{\mathbf{x}}^{(2)}\|^2$
6. Update $\text{Enc}^{(1)}, \text{Dec}, \text{RE-Attn}, \text{Enc}^{(2)}$ via $\nabla \mathcal{L}_{\text{recon}}$

Phase 2 — Latent Discriminator Update

1. Sample $\mathbf{z}_{\text{real}} \sim \mathcal{N}(0, I_4)$
2. $\mathbf{z}_{\text{enc}} \leftarrow \text{Enc}^{(2)}(\tilde{\mathbf{x}})$ *// encoder output (frozen here)*
3. $\mathcal{L}_D \leftarrow -\mathbb{E}[\log D_{\text{lat}}(\mathbf{z}_{\text{real}})] - \mathbb{E}[\log(1 - D_{\text{lat}}(\mathbf{z}_{\text{enc}}))]$
4. Update D_{lat} via $\nabla \mathcal{L}_D$

Phase 3 — Encoder Adversarial Update

1. $\mathcal{L}_G \leftarrow -\mathbb{E}[\log D_{\text{lat}}(\text{Enc}^{(2)}(\tilde{\mathbf{x}}))]$
2. Update $\text{Enc}^{(2)}$ via $\nabla \mathcal{L}_G$ *// encoder learns to fool D_{lat}*

5.4. Anomaly Scoring

At test time, for each sample $\mathbf{x}$, two scores are computed:

$$s_{\text{recon}} = \frac{1}{d} \|\mathbf{x} - \hat{\mathbf{x}}\|^2 \quad (8)$$

$$s_{\text{latent}} = 1 - D_{\text{lat}}(\mathbf{z}^{(2)}) \quad (9)$$

$$s_{\text{final}}(\alpha) = \alpha \cdot s_{\text{recon}} + (1 - \alpha) \cdot s_{\text{latent}}, \quad \alpha \in [0, 1] \quad (10)$$

The reconstruction score s_{recon} captures attacks that distort many features simultaneously (e.g., DoS). The latent score s_{latent} captures subtle attacks whose latent code falls outside the Gaussian prior even if individual features appear normal (e.g., R2L, U2R). The α parameter is swept to find the optimal blend, yielding an interpretable curve of AUC-ROC vs. α .

6. Experimental Design

6.1. Ablation Conditions

The experiment is a five-condition ablation study where each condition adds exactly one architectural component. This isolates the contribution of each component independently. The difference $C3 \rightarrow C4$ tests the value of the dual scoring mechanism. The difference $C4 \rightarrow C5$ tests the specific novel contribution: **error-guided attention**.

6.2. Evaluation Metrics

For each condition the following metrics are computed on the test set:

- **AUC-ROC**: area under the receiver operating characteristic curve; measures overall ranking quality.
- **AUC-PR**: area under the precision-recall curve; more informative for the highly imbalanced 1% anomaly setting.
- **F1-score**: at the threshold that maximises F1 on the test set.
- **False Negative Rate (FNR)**: proportion of attacks missed; critical for security applications.
- **Per-family AUC**: AUC-ROC computed separately for DoS, Probe, R2L, and U2R to reveal which attack families benefit most from error-guided attention.

6.3. Additional Analysis: α -Sweep

For conditions C4 and C5, the blend parameter α is swept from 0 to 1 in steps of 0.05. A plot of AUC-ROC vs. α for each condition directly reveals:

- The optimal α for each condition.
- Whether the dual score is additive (peak strictly between 0 and 1) or dominated by one signal.
- Whether RE-Attention changes the optimal blend point, indicating that it creates a qualitatively different latent representation.

Table 3: Five Experimental Conditions and Their Architectural Components

C#	Name	RE-Attn	Adv. Prior	Dual Score	Tests
C1	Vanilla AE [Zavrak & Iskefiyeli, 2021]	×	×	×	Reconstruction-only lower bound
C2	GAN discriminator [Zenati et al., 2018]	×	×	×	Discriminator-only lower bound
C3	AAE (no attention)	×	✓	Recon only	Does Gaussian prior improve reconstruction-based detection?
C4	AAE + Dual Score	×	✓	✓	Does the latent score add signal beyond reconstruction?
C5	RE-Attn-AAE (full)	✓	✓	✓	Does error-guided attention further improve C4?

6.4. Interpretability Analysis: Attention Weight Visualisation

After training C5, the attention weights $\mathbf{a}$ are recorded for *all correctly detected test anomalies* grouped by attack family. The mean attention weight per feature per family is plotted as a heatmap (114 features $\times$ 4 attack families). This answers the question: *Which network traffic features does the model focus on when it detects each attack type?*

This is the key interpretability result. Existing attention works [Abdulganiyu et al., 2025, Al-Qatf et al., 2021] do not provide this feature-level analysis on KDD99/NSL-KDD attack families.

6.5. Hyperparameters

Table 4: Shared Hyperparameters Across All Conditions

Parameter	Value
Batch size	512
Optimiser	Adam ($\beta_1 = 0.5$, $\beta_2 = 0.999$)
Learning rate	1×10^{-5}
Epochs	30 (C1–C5)
Latent dimension (d_z)	4
RE-Attention hidden size	64
Latent discriminator	$4 \rightarrow 32 \rightarrow 16 \rightarrow 1$
Anomaly % in training	1%
Random seed	42

6.6. Infrastructure

All experiments run in a single Kaggle notebook on a T4 GPU (`tensorflow==2.x`, `scikit-learn`, `numpy`). Estimated total runtime: ≈ 35 minutes for all five conditions at 30 epochs, well within Kaggle’s 9-hour session limit and the 2-hour target.

7. Expected Results and Hypotheses

- H1. Adversarial prior improves over vanilla AE (C3 vs C1):** Enforcing a Gaussian prior on the latent space should improve AUC-PR because anomalous latent codes are more likely to fall outside the regularised distribution, even before adding the explicit latent score. We expect $C3 \text{ AUC-PR} > C1 \text{ AUC-PR}$.
- H2. Dual scoring adds complementary signal (C4 vs C3):** The reconstruction score alone is most effective for DoS attacks (which distort many features). The latent score should add detection power for Probe and R2L attacks (which are subtler in feature space). We expect a distinct peak in the AUC-ROC vs. α curve at $0 < \alpha^* < 1$ for C4, confirming that both signals contribute.
- H3. RE-Attention further improves over input-driven alternatives (C5 vs C4):** Error-guided attention directs the second-pass encoder to concentrate on the most anomaly-relevant features. We expect this to improve per-family AUC for the rarer and harder attack families (R2L, U2R), where the anomaly signature is concentrated in a small number of features that the error vector will highlight.
- H4. Attention weights are interpretable and consistent with domain knowledge:** We expect the heatmap to show high attention on `error_rate`, `count`, `src_bytes` for DoS; on `dst_host_count`, `dst_host_srv_count` for Probe; on `num_failed_logins`,

is_guest_login for R2L; and on root_shell, num_root for U2R. Agreement with known network attack signatures validates the attention mechanism.

8. Limitations and Risks

- **KDD99 is a synthetic dataset:** It was generated from a simulated military network in 1998. Results may not transfer directly to modern network traffic. NSL-KDD reduces but does not eliminate this concern.
- **Latent dimension:** The 4-dimensional bottleneck is very small for the dual encoder design. If the second encoder does not converge stably, the latent discriminator may receive noisy gradients. A larger latent dimension (8 or 16) is a contingency.
- **Three-phase training instability:** Alternating three gradient updates per batch is standard for adversarial AEs but can be sensitive to the relative learning rates of encoder and discriminator. Careful monitoring of discriminator loss is required.
- **Dual encoder parameter count:** The full RE-Attn-AAE has approximately double the encoder parameters of the vanilla AE baseline (C1). This is controlled for in the ablation by comparing all conditions at the same number of training epochs.

9. Conclusion

This proposal describes a focused architectural experiment for network anomaly detection on the KDD99 and NSL-KDD datasets. The proposed **RE-Attention Adversarial Autoencoder** advances beyond established reconstruction-based [Zavrak & Iskefiyeli, 2021] and adversarial [Zenati et al., 2018, Lunardi et al., 2022] baselines through three principled contributions: adversarial Gaussian regularisation of the latent space, dual anomaly scoring combining reconstruction and latent normality signals, and a novel error-guided attention mechanism that uses the first-pass reconstruction failure to re-weight the input for a more targeted second-pass encoding.

A literature search confirmed that while adversarial AEs, attention AEs, and dual scoring have been explored individually and in partial combinations [Zenati et al., 2018, Lunardi et al., 2022, Benabderrahmane et al., 2025, Puuska et al., 2018, Abdulganiyu et al., 2025, Al-Qatf et al., 2021], the specific mechanism of computing attention weights from the per-feature reconstruction error vector has not been reported for network intrusion detection. This constitutes the novel contribution of the experiment.

The controlled five-condition ablation study allows each component’s contribution to be measured independently. The interpretability analysis — mapping attention weights to KDD99 feature names and attack families — provides a qualitative result that complements the quantitative metrics and may offer genuine insight into which aspects of network traffic most strongly signal each attack type.

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